

Cointeractions of Some Word Bialgebras

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Definition

An algebra A over a field K is a vector space over K with two linear maps, the multiplication map $m : A \otimes A \rightarrow A$ and the unit map $u : K \rightarrow A$ such that the following two diagrams commute.

$$\begin{array}{ccc}
 A \otimes A \otimes A & \xrightarrow{\text{Id} \otimes m} & A \otimes A \\
 \downarrow m \otimes \text{Id} & & \downarrow m \\
 A \otimes A & \xrightarrow{m} & A
 \end{array}$$

$$\begin{array}{ccccc}
 K \otimes A & \xleftarrow{a \mapsto 1 \otimes a} & A & \xrightarrow{a \mapsto a \otimes 1} & A \otimes K \\
 \downarrow u \otimes \text{Id} & & \downarrow \text{Id} & & \downarrow \text{Id} \otimes u \\
 A \otimes A & \xrightarrow{m} & A & \xleftarrow{m} & A \otimes A
 \end{array}$$

Clarification on the Multiplication

Note that the multiplication map here is defined on $A \otimes A$ instead of the usual $A \times A$. This definition is valid because of the universal property of tensor products.

Theorem (Universal Property of Tensor Product)

For any algebras A, B and the bilinear map $f : A \times A \rightarrow B$, there exists a unique linear map $\hat{f} : A \otimes A \rightarrow B$ such that $r \circ \hat{f} = f$.

Here r is the bilinear map $r : A \otimes A \rightarrow A \otimes A$ such that $r(a_1, a_2) = a_1 \otimes a_2$ for any $a_1, a_2 \in A$.

$$\begin{array}{ccc} A \times A & \xrightarrow{r} & A \otimes A \\ & \searrow f & \downarrow \hat{f} \\ & & B \end{array}$$

Definition

A coalgebra C over a field K is a vector space over K with two linear maps, the coproduct $\Delta : C \rightarrow C \otimes C$, and the counit map $\epsilon : C \rightarrow K$ such that the following two diagrams commute.

$$\begin{array}{ccc}
 C \otimes C \otimes C & \xleftarrow{\text{Id} \otimes \Delta} & C \otimes C \\
 \Delta \otimes \text{Id} \uparrow & & \uparrow \Delta \\
 C \otimes C & \xleftarrow{\Delta} & C
 \end{array}$$

$$\begin{array}{ccccc}
 K \otimes C & \xrightarrow{k \otimes c \mapsto kc} & C & \xleftarrow{c \otimes k \mapsto kc} & C \otimes K \\
 \epsilon \otimes \text{Id} \uparrow & & \text{Id} \uparrow & & \text{Id} \otimes \epsilon \uparrow \\
 C \otimes C & \xleftarrow{\Delta} & C & \xrightarrow{\Delta} & C \otimes C
 \end{array}$$

Algebra Homomorphism

Definition

Let A and B be algebras over K . Then the map $f : A \rightarrow B$ is an algebra homomorphism if the following two diagrams commute.

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ m_A \uparrow & & \uparrow m_B \\ A \otimes A & \xrightarrow{f \otimes f} & B \otimes B \end{array}$$

$$\begin{array}{ccc} K & \xrightarrow{u_B} & B \\ u_A \downarrow & \nearrow f & \\ A & & \end{array}$$

Coalgebra Homomorphism

Definition

Let C and D be coalgebras over K . Then the map $g : C \rightarrow D$ is a coalgebra homomorphism if the following two diagrams commute.

$$\begin{array}{ccc} C & \xrightarrow{g} & D \\ \downarrow \Delta_C & & \downarrow \Delta_D \\ C \otimes C & \xrightarrow{g \otimes g} & D \otimes D \end{array}$$

$$\begin{array}{ccc} C & \xrightarrow{g} & D \\ & \searrow \epsilon_C & \downarrow \epsilon_D \\ & & K \end{array}$$

Theorem

If a vector space B is simultaneously an algebra and a coalgebra over K , then the product and unit are coalgebra homomorphisms if and only if the coproduct and counit are algebra homomorphisms.

Definition

If the vector space B is both an algebra and a coalgebra over K and the coproduct and counit are algebra homomorphisms, then B is a bialgebra over K .

$m_{1,3,2,4}$ Map I

If the coproduct is an algebra homomorphism, then the following four diagrams commute. Note that the unit map from K to K is the identity map.

$$\begin{array}{ccc}
 B & \xrightarrow{\Delta} & B \otimes B \\
 m \uparrow & & \uparrow m_{13,24} \\
 B \otimes B & \xrightarrow{\Delta \otimes \Delta} & B \otimes B \otimes B \otimes B
 \end{array}$$

$$\begin{array}{ccc}
 K & \xrightarrow{u_{B \otimes B}} & B \otimes B \\
 u \downarrow & \nearrow \Delta & \\
 B & &
 \end{array}$$

$$\begin{array}{ccc}
 B & \xrightarrow{\epsilon} & K \\
 m \uparrow & & \uparrow m_K \\
 B \otimes B & \xrightarrow{\epsilon \otimes \epsilon} & K \times K
 \end{array}$$

$$\begin{array}{ccc} K & \xrightarrow{\text{Id}} & K \\ u \downarrow & \nearrow \epsilon & \\ B & & \end{array}$$

We introduce the map $m_{1,3,2,4} : A \otimes A \otimes A \otimes A \rightarrow A \otimes A \otimes A \otimes A$ such that

$$m_{1,3,2,4}(a_1 \otimes a_2 \otimes a_3 \otimes a_4) = a_1 \otimes a_3 \otimes a_2 \otimes a_4,$$

and let $m_{13,24} = (m \otimes m) \circ m_{1,3,2,4}$.

Concatenation-Deshuffle Bialgebra I

Let $\hat{m}$ be the concatenation of words and i be the inclusion map. By the universal property of vector spaces, there exists a unique linear map m which we define as the multiplication for the bialgebra. We call this multiplication concatenation.

$$\begin{array}{ccc} \mathcal{W} \otimes \mathcal{W} & \xrightarrow{i} & W \otimes W \\ & \searrow \hat{m} & \downarrow m \\ & & W \end{array}$$

Similarly, we define coproduct $\hat{\Delta}$ over the the basis $\mathcal{W}$. For any non-empty word $a_1 \cdots a_n \in \mathcal{W}$, where $a_1, \dots, a_n \in \Omega$, $n \in \mathbb{Z}_+$, we let the $\hat{\Delta}$ be the deshuffle map such that

$$\hat{\Delta}(a_1 \cdots a_n) = \sum_{I \subset \{1, \dots, n\}} a_I \otimes a_{\{1, \dots, n\} \setminus I},$$

Concatenation-Deshuffle Bialgebra II

where a_I is a subword of $a_1 \cdots a_n$ consisting of letters indexed by I . We define $\hat{\Delta}(\mathcal{E}) = \mathcal{E} \otimes \mathcal{E}$, where $\mathcal{E}$ is the empty word. For example, we have

$$\hat{\Delta}(aba) = \mathcal{E} \otimes aba + a \otimes ba + b \otimes aa + a \otimes ab + ab \otimes a + aa \otimes b + ba \otimes a + aba \otimes \mathcal{E},$$

where $a, b \in \Omega$. By the universal property of vector spaces, there exists a unique linear map Δ which we define as the coproduct for the bialgebra. We call this coproduct deshuffle.

$$\begin{array}{ccc} \mathcal{W} & \xrightarrow{i} & W \\ & \searrow \hat{\Delta} & \downarrow \Delta \\ & & W \otimes W \end{array}$$

For any $k \in K$, we can define the unit map $u(k) = k\mathcal{E}$ where $\mathcal{E}$ is the empty word.

Concatenation-Deshuffle Bialgebra III

We define the counit $\hat{\epsilon}$ over the basis $\mathcal{W}$. Let $\hat{\epsilon}(\mathcal{E}) = 1$ and $\hat{\epsilon}(w) = 0$ for any non-empty word $w \in \mathcal{W}$. By the universal property of vector spaces, there exists a unique linear map ϵ which we define as the counit for the bialgebra.

$$\begin{array}{ccc} \mathcal{W} & \xrightarrow{i} & W \\ & \searrow \hat{\epsilon} & \downarrow \epsilon \\ & & K \end{array}$$

Shuffle-Deconcatenation Bialgebra I

Similar to our discussion above, by the universal property of vector spaces, it suffices to define the multiplication map $\hat{m}$ over the basis $\mathcal{W} \otimes \mathcal{W}$. For any $w_1, w_2 \in \mathcal{W}$, let

$\hat{m}(w_1, w_2)$ = sum of all words formed by shuffling w_1 and w_2 . If $w_1 = \mathcal{E}$, then $\hat{m}(w_1, w_2) = w_2$ and if $w_2 = \mathcal{E}$, then $\hat{m}(w_1, w_2) = w_1$. If $w_1 = u_1 \cdots u_k, w_2 = u_{k+1} \cdots u_{k+l} \in \mathcal{W}$ where $k, l \in \mathbb{Z}_+$, then

$$\hat{m}(w_1, w_2) = \sum_{\sigma \text{ permutes } \{1, \dots, k+l\}} u_{\sigma(1)} \cdots u_{\sigma(k+l)},$$

where $\sigma^{-1}(1) < \cdots < \sigma^{-1}(k)$ and $\sigma^{-1}(k+1) < \cdots < \sigma^{-1}(k+l)$.

Shuffle-Deconcatenation Bialgebra II

We have the following examples,

$$uv \sqcup x = uvx + uxv + xuv,$$

$$\begin{aligned} uv \sqcup ux &= uvux + uuvx + uuvx + uuxv + uuxv + uxuv \\ &= uvux + 2uuvx + 2uuxv + uxuv. \end{aligned}$$

Shuffle-Deconcatenation Bialgebra III

Next we define the coproduct over the basis $\mathcal{W}$. Let $\hat{\Delta}(\mathcal{E}) = \mathcal{E} \otimes \mathcal{E}$. For any $w = u_1 \cdots u_k$ where $k \in \mathbb{Z}_+$, we have

$$\hat{\Delta}(w) = \mathcal{E} \otimes w + w \otimes \mathcal{E} + \sum_{i=1}^{k-1} u_1 \cdots u_i \otimes u_{i+1} \cdots u_k.$$

We call the coproduct from the universal property of vector spaces deconcatenation, and we have the following examples,

$$\Delta(aba) = \mathcal{E} \otimes aba + aba \otimes \mathcal{E} + a \otimes ba + ab \otimes a,$$

where $a, b \in \Omega$.

We use the same definitions of unit and counit in the concatenation-deshuffle bialgebra.

Definition

A vector space V over a field K is graded (or $\mathbb{Z}_{\geq 0}$ -graded to be more precise) if it has a direct sum decomposition $V = \bigoplus_{i=0}^{\infty} V_i$, where $V_1, V_2, \dots$ are vector spaces over K . For any $i \in \mathbb{Z}_{\geq 0}$, the vector space V_i is called the graded piece of degree i and the elements of V_i are called homogeneous of degree i .

We observe that if V is a graded vector space, then $V \otimes V$ is also graded. Specifically, the graded piece of degree n is

$$(V \otimes V)_n = \bigoplus_{j=0}^n V_j \otimes V_{n-j}$$

for any $n \in \mathbb{Z}_{\geq 0}$.

Let $\mathcal{W}_n$ be the set of word over the alphabet Ω with length n where $n \in \mathbb{Z}_{\geq 0}$, and let $W_n = \text{span}_K \mathcal{W}_n$. We have $W = \bigcup_{i=0}^{\infty} \mathcal{W}_n$ and we can see that $W = \bigoplus_{i=0}^{\infty} W_n$, so W is graded with W_i being the graded piece of degree i for any $i \in \mathbb{Z}_{\geq 0}$.

Definition

A graded vector space V over K is connected if $V_0 \cong K$.

Since $W_0 = \text{span}_K \{\mathcal{E}\} \cong K$, we can see that W is a connected graded vector space.

Definition

A linear map $f : V \rightarrow W$, where V and W are graded vector spaces, is graded if $f(V_n) \subset W_n$ for any $n \in \mathbb{Z}_{\geq 0}$.

Definition

An algebra, coalgebra, or bialgebra is graded if the underlying vector space and all the defining maps are graded.

For any $n \in \mathbb{Z}_{\geq 0}$, the graded piece of degree n of $W \otimes W$ is $\bigoplus_{j=0}^n W_j \otimes W_{n-j}$. With our definition of concatenation-deshuffle and shuffle-deconcatenation bialgebras, we can see that both multiplications map the graded piece of degree n of $W \otimes W$ to the graded piece of degree n of W , and both coproducts map the graded piece of degree n of W to the graded piece of degree n of $W \otimes W$. Therefore, both bialgebra are graded.

Definition (Convolution)

For any algebra A and coalgebra C , let $f, g : C \rightarrow A$ be linear maps. The convolution product of f and g is

$$f * g = m \circ (f \otimes g) \circ \Delta.$$

Definition

A bialgebra B is a Hopf algebra if there exists a linear map $S : B \rightarrow B$, which we call the antipode, such that the following diagram commutes.

$$\begin{array}{ccccc}
 & & B \otimes B & \xrightarrow{S \otimes \text{Id}} & B \otimes B \\
 & \nearrow \Delta & & & \downarrow m \\
 B & \xrightarrow{\epsilon} & K & \xrightarrow{u} & B \\
 & \searrow \Delta & & & \uparrow m \\
 & & B \otimes B & \xrightarrow{\text{Id} \otimes S} & B \otimes B
 \end{array}$$

By the definition of convolution product, we can see that this diagram shows that the antipode has to satisfy the relation $S * \text{Id} = \text{Id} * S = u \circ \epsilon$.

Theorem

- 1 For any Hopf algebra A , the antipode S is an algebra anti-automorphism, i.e., S is an algebra isomorphism, $S(\mathcal{E}) = \mathcal{E}$ and $S(ab) = S(b)S(a)$ for any $a, b \in A$.
- 2 For any Hopf algebra A , if A is commutative or cocommutative then $S \circ S = \text{Id}$.
- 3 For any graded connected bialgebra A , A has a unique antipode S which is determined recursively. In addition, S is a graded map, so A is a graded Hopf algebra.

The recursive relation is

$$S(x) = -x - \sum_i S(x_{i,1})x_{i,2}.$$

By induction, we can show that the antipode for both bialgebra is

$$S(w_1 \cdots w_k) = (-1)^k w_k \cdots w_1,$$

for any $w_1 \cdots w_k \in W$ where $k \in \mathbb{Z}_{\geq 0}$.

Definition

Let $(P, \leq)$ be a partially ordered set (poset). An interval in P is a nonempty subposet of the form

$$[x, y] = \{z \in P \mid x \leq z \leq y\},$$

for any $x, y \in P$. We denote the set of all intervals in P as $\text{int}(P)$.

Definition

A poset is locally finite if all its intervals are finite.

Definition

Let P be a locally finite poset and $C = \text{span}_K \text{int}(P)$ be the vector space over K constructed from the basis $\text{int}(P)$. The incidence coalgebra of P is the coalgebra C with the coproduct

$$\Delta([x, y]) = \sum_{z \in [x, y]} [x, z] \otimes [z, y],$$

and the counit

$$\epsilon([x, y]) = \begin{cases} 1 & \text{if } x = y \\ 0 & \text{if } x \neq y \end{cases},$$

for any $[x, y] \in \text{int}(P)$.

Concatenation-Deshuffle Bialgebra on Two Disjoint Alphabets I

Let $\mathcal{W}$ be the set of all words over the alphabet $\Omega \cup \Omega'$, and let $W = \text{span}_K \mathcal{W}$ be the vector space of formal linear combinations of words over the field K . By the definition of W , we can see that $\mathcal{W}$ is a basis of W .

Consider the map $f : \Omega \rightarrow \mathcal{P}(\Omega')$, where $\mathcal{P}(\Omega') = \{S \mid S \subset \Omega'\}$ is the power set of Ω' , which maps each letter in Ω to the set of letters it corresponds to in Ω' . Note that for each $a \in \Omega$ in w , the letters of $f(a)$ also appear in w , and we choose particular copies of the letters of $f(a)$ and consider that they are associated with that particular a .

For example, let $\Omega = \{a_1, a_2\}$ and $\Omega' = \{b_1, b_2, b_3\}$. Let $f(a_1) = \{b_1, b_2\}$ and $f(a_2) = \{b_1, b_2, b_3\}$. We consider the word $w = a_1 b_1 b_2 a_2 b_1 b_3 \in \mathcal{W}$, where a_1 corresponds to b_1, b_2 of indices 2, 3 and a_2 corresponds to b_1, b_2, b_3 of indices 5, 3, 6 (Note that

Concatenation-Deshuffle Bialgebra on Two Disjoint Alphabets II

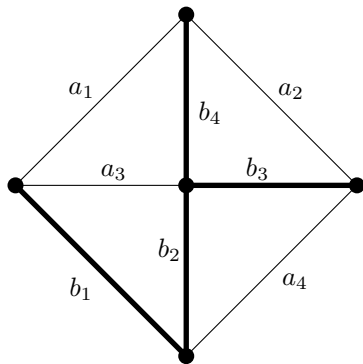
the same letter in Ω' inside a word can be associated with multiple different letters in Ω inside that word; here b_2 of index 3 is associated with both a_1 of index 1 and a_2 of index 4). The map f is insufficient to determine the exact correspondence relation within each word. We can see that there are two different b_1 's in w . If a_1 corresponds to b_1, b_2 of indices 5, 3 and a_2 corresponds to b_1, b_2, b_3 of indices 2, 3, 6, we have a difference correspondence relation that also satisfies f .

For any $w = w_1 \cdots w_k \in W$ where $k \in \mathbb{Z}_+$, we introduce the map $\tilde{f}_w : \{1, \dots, k\} \rightarrow \mathcal{P}(\{1, \dots, k\})$ to make uniquely determine the correspondence relation within the word w . For any $i = 1, \dots, k$, if $w_i \in \Omega$, let $\tilde{f}_w(i)$ be the set of indices of the one copy of elements of $f(w_i)$ that w_i corresponds to. If $w_i \notin \Omega$, let $\tilde{f}_w(i) = \emptyset$.

Concatenation-Deshuffle Bialgebra on Two Disjoint Alphabets III

For the word above $w = a_1 b_1 b_2 a_2 b_1 b_3 \in \mathcal{W}$, where a_1 corresponds to b_1, b_2 of indices 2, 3 and a_2 corresponds to b_1, b_2, b_3 of indices 5, 3, 6, we have $\tilde{f}_w(1) = \{2, 3\}$, $\tilde{f}_w(4) = \{3, 5, 6\}$, and $\tilde{f}_w(2) = \tilde{f}_w(3) = \tilde{f}_w(5) = \tilde{f}_w(6) = \emptyset$.

Concatenation-Deshuffle Bialgebra on Two Disjoint Alphabets IV



For the graph G on the left, we fix a spanning tree T . We associate each edge in T with a letter in Ω' and each edge not in T with a letter in Ω . Since T is the spanning tree of G , adding an additional edge e would create a unique cycle C . We call this cycle

Concatenation-Deshuffle Bialgebra on Two Disjoint Alphabets V

C the fundamental cycle of the edge e . For any edge e not in the spanning tree T , we want $\{e\} \cup f(e)$ to be the fundamental cycle of e . Hence, we have $f(a_1) = \{b_1, b_2, b_4\}$, $f(a_2) = \{b_3, b_4\}$, $f(a_3) = \{b_1, b_2\}$, and $f(a_4) = \{b_2, b_3\}$.

The coproduct Δ deshuffles the word with respect to the alphabet Ω and puts letters in the alphabet Ω' in the result in a way such that the correspondence relation and the relative position are preserved. For example, for the word $w = a_1 b_1 b_2 a_2 b_1 b_3$ described above, we have

$$\begin{aligned}\Delta(w) = & a_1 b_1 b_2 a_2 b_1 b_3 \otimes \mathcal{E} + \mathcal{E} \otimes a_1 b_1 b_2 a_2 b_1 b_3 \\ & + a_1 b_1 b_2 \otimes b_2 a_2 b_1 b_3 + b_2 a_2 b_1 b_3 \otimes a_1 b_1 b_2\end{aligned}$$

Concatenation-Deshuffle Bialgebra on Two Disjoint Alphabets VI

We have $\Delta(\mathcal{E}) = \mathcal{E} \otimes \mathcal{E}$, and for any $w \in \mathcal{W}_f$ in which there are n letters in Ω for any $n \in \mathbb{Z}_+$, we have

$$\Delta(w) = \sum_{I \subset \{1, \dots, n\}} w_I^* \otimes w_{\{1, \dots, n\} \setminus I}^*,$$

where w_I^* is the word formed from w by choosing all i -th letters in Ω for $i \in I$ and filling in letters from Ω' in a way such that the correspondence relation and the relative position are preserved. We use the same definitions of unit and counit in the concatenation-deshuffle bialgebra on one alphabet.

Concatenation-Deshuffle Bialgebra on Two Disjoint Alphabets VII

Furthermore, we want to check whether W_f is a Hopf algebra. We could check whether the antipode $S(w) = (-1)^k w^R$ for any $w \in W_f$ would work, where k is the length of w and W^R is the reverse of the word w . Let $w = a_1 b_1 a_2$, where $\tilde{f}_w(1) = \tilde{f}_w(3) = \{2\}$ and $\tilde{f}_w(2) = \emptyset$. We have

$$\Delta(w) = a_1 b_1 a_2 \otimes \mathcal{E} + \mathcal{E} \otimes a_1 b_1 a_2 + a_1 b_1 \otimes b_1 a_2 + b_1 a_2 \otimes a_1 b_1,$$

so

$$(S \otimes \text{Id}) \circ \Delta(w) = -a_2 b_1 a_1 \otimes \mathcal{E} + \mathcal{E} \otimes a_1 b_1 a_2 + b_1 a_1 \otimes b_1 a_2 + a_2 b_1 \otimes a_1 b_1,$$

and thus

$$S * \text{Id}(w) = -a_2 b_1 a_1 + a_1 b_1 a_2 + b_1 a_1 b_1 a_2 + a_2 b_1 a_1 b_1.$$

Concatenation-Deshuffle Bialgebra on Two Disjoint Alphabets VIII

We also have $u \circ \epsilon(w) = u(\epsilon(w)) = 0 \neq (S \otimes \text{Id}) \circ \Delta(w)$, so S is not an antipode of W_f . The main reason why this definition fails is that the concatenation product of each summand can be of different length when a letter in Ω' is associated with multiple letters in Ω .

Concatenation-Deshuffle Bialgebra on Two Disjoint Alphabets IX

Unlike the concatenation-deshuffle bialgebra on one alphabet, W_f is not graded on the length of words. If we were to grade the bialgebra on the length of words, $w = a_1 b_1 a_2$ described above would be homogeneous of degree 3. However, the coproduct $\Delta(w) = a_1 b_1 a_2 \otimes \mathcal{E} + \mathcal{E} \otimes a_1 b_1 a_2 + a_1 b_1 \otimes b_1 a_2 + b_1 a_2 \otimes a_1 b_1$ is not in any graded piece, so the coproduct map would not be graded. As we have discussed before, this situation happens when a letter in Ω' is associated with multiple letters in Ω .

Since the coproduct is based on deshuffling with respect to Ω , one natural way to fix this issue is to try to grade the bialgebra on the number of letters in Ω . Let $\mathcal{W}_{f,n}$ be the set of words in $\mathcal{W}_f$ with n letters in Ω , and let $W_{f,n} = \text{span}_K \mathcal{W}_{f,n}$. We can see that $W_f = \bigoplus_{n=0}^{\infty} W_{f,n}$, so the vector space W_f is graded. Specifically, we can see that $\mathcal{W}_{f,0} = \{\mathcal{E}\}$, so $W_{f,0} \cong K$.

Concatenation-Deshuffle Bialgebra on Two Disjoint Alphabets X

The graded piece of degree n for $W_f \otimes W_f$ is $\bigoplus_{j=0}^n W_{f,j} \otimes W_{f,n-j}$ for any $n \in \mathbb{Z}_{\geq 0}$. For any $w_1 \otimes w_2 \in W_{f,i} \otimes W_{f,j}$ of degree $(i+j)$, where $i, j \in \mathbb{Z}_{\geq 0}$, the concatenation product $m(w_1 \otimes w_2)$ has $(i+j)$ letters in Ω and is of degree $(i+j)$. Hence, the concatenation product map is graded. We can see that $\Delta(\mathcal{E}) = \mathcal{E} \otimes \mathcal{E}$ where both $\mathcal{E}$ and $\mathcal{E} \otimes \mathcal{E}$ are of degree 0. For any $w \in W_{f,n}$ of degree n , where $n \in \mathbb{Z}_+$, we have

$$\Delta(w) = \sum_{I \subset \{1, \dots, n\}} w_I^* \otimes w_{\{1, \dots, n\} \setminus I}^*.$$

By definition w_I^* is of degree $|I|$ and $w_{\{1, \dots, n\} \setminus I}^*$ is of degree $(n - |I|)$, so any summand $w_I^* \otimes w_{\{1, \dots, n\} \setminus I}^*$ is of degree n (here $|I|$ is the cardinality of I). Hence, the coproduct is graded. We conclude that W_f is a graded connected bialgebra.

Concatenation-Deshuffle Bialgebra on Two Disjoint Alphabets XI

For any $w \in W_f$ with n letters in Ω where $n \in \mathbb{Z}_+$, we have $\epsilon(w) = 0$, so $w \in \ker \epsilon$. We have

$$S(w) = -w - \sum_i S(w_{i,1})w_{i,2}$$

where $\tilde{\Delta}(w) = \sum_i w_{i,1} \otimes w_{i,2}$ is the non-primitive part of $\Delta(w)$. Consider the example $w = a_1 b_1 a_2$, where $\tilde{f}_w(1) = \tilde{f}_w(3) = \{2\}$ and $\tilde{f}_w(2) = \emptyset$. We have

$$\begin{aligned} S(w) &= -w - S(a_1 b_1) b_1 a_2 - S(b_1 a_2) a_1 b_1 \\ &= -a_1 b_1 a_2 + a_1 b_1 b_1 a_2 + b_1 a_2 a_1 b_1. \end{aligned}$$

As we can see, $S(w)$ has no clear relation to w , so there is no clean expression for antipode for W_f .

Incidence Structure on the Concatenation-Deshuffle Bialgebra on Two Disjoint Alphabets I

For any word $w \in \mathcal{W}$, let I_w be the set of indices of letters in Ω' . We can see that $\mathcal{P}(I_w)$ is a poset. Since w is of finite length, I_w is also finite, so any interval in $\mathcal{P}(I_w)$ and thus $\mathcal{P}(I_w)$ is locally finite. Let

$$WI = \bigoplus_{w \in \mathcal{W}_f} \text{span}_K \{(w, [A, B]) \mid [A, B] \in \text{int}(\mathcal{P}(I_w))\},$$

and $C_w = \text{span}_K \text{int}(\mathcal{P}(I_w))$, and we can see that $WI \cong \bigoplus_{w \in \mathcal{W}_f} C_w$. We can see that WI is a vector space over K with the basis

$$\mathcal{B} = \bigcup_{w \in \mathcal{W}_f} \{(w, [A, B]) \mid [A, B] \in \text{int}(\mathcal{P}(I_w))\}.$$

Incidence Structure on the Concatenation-Deshuffle Bialgebra on Two Disjoint Alphabets II

We can introduce an incidence coalgebra structure on WI with the coproduct

$$\delta(w, [A, B]) = \sum_{A \subset C \subset B} (w, [A, C]) \otimes (w, [C, B]),$$

and the counit

$$\epsilon_{\delta}(w, [A, B]) = \begin{cases} 1 & \text{if } A = B \\ 0 & \text{if } A \neq B \end{cases},$$

for any $(w, [A, B]) \in \mathcal{B}$.

Incidence Structure on the Concatenation-Deshuffle Bialgebra on Two Disjoint Alphabets III

Instead of using the multiplication map from the incidence algebra structure, we want to define the multiplication map in the following way.

$$m((w_1, [A, B]) \otimes (w_2, [C, D])) = (\overline{w_1 w_2}, [A \cup C^*, B \cup D^*]),$$

for any $w_1, w_2 \in \mathcal{W}_f$, $[A, B] \in \text{int}(\mathcal{P}(I_{w_1}))$, and $[C, D] \in \text{int}(\mathcal{P}(I_{w_2}))$, where $C^* = C + |w_1|$, $D^* = D + |w_1|$, and $|w_1|$ is the length of w_1 .

For example, let $w_1 = a_1 b_1 b_2$ and $w_2 = b_1 a_2 b_3$, and we have

$$\begin{aligned} & m((w_1, [\{3\}, \{2, 3\}]) \otimes (w_2, [\{1\}, \{1, 3\}])) \\ &= (a_1 b_1 b_2 b_1 a_2 b_3, [\{3, 4\}, \{2, 3, 4, 6\}]). \end{aligned}$$

Incidence Structure on the Concatenation-Deshuffle Bialgebra on Two Disjoint Alphabets IV

We want to make sure that $m : V \otimes V \rightarrow V$ has the property that the underlying vector space V is the same as WI . We can see that the basis of V is

$$\begin{aligned} & \bigcup_{w \in \mathcal{W}_f} \bigcup_{[A, B] \in \text{int}(\mathcal{P}(I_w))} \{(w, [A, B])\} \\ &= \bigcup_{w \in \mathcal{W}_f} \{(w, [A, B]) \mid [A, B] \in \text{int}(\mathcal{P}(I_w))\} = \mathcal{B}, \end{aligned}$$

so $V = WI$.

Incidence Structure on the Concatenation-Deshuffle Bialgebra on Two Disjoint Alphabets V

We can also introduce another coproduct Δ based on the deshuffle coproduct of $\mathcal{W}_f$. For any $w \in \mathcal{W}_f$ and $[A, B] \in \text{int}(\mathcal{P}(I_w))$, we define the coproduct on $\mathcal{B}$

$$\hat{\Delta}(w, [A, B]) = \sum_i (w_{i,1}, [A_{w_{i,1}}, B_{w_{i,1}}]) \otimes (w_{i,2}, [A_{w_{i,2}}, B_{w_{i,2}}]),$$

where $\Delta(w) = \sum_i w_{i,1} \otimes w_{i,2}$ for the coproduct on the concatenation-deshuffle bialgebra on two alphabets, and $A_{w_{i,1}}$ (similarly for $B_{w_{i,1}}$, $A_{w_{i,2}}$ and $B_{w_{i,2}}$) is A restricted to letters in Ω' in $w_{i,1}$ with indices adjusted accordingly.

Incidence Structure on the Concatenation-Deshuffle Bialgebra on Two Disjoint Alphabets VI

For example, let $w = a_1 b_1 b_2 a_2 b_3 \in \mathcal{W}_f$ where $\tilde{f}_{w_2}(1) = \{2, 3\}$, $\tilde{f}_{w_2}(4) = \{2, 5\}$, and $\tilde{f}_{w_2}(2) = \tilde{f}_{w_2}(3) = \tilde{f}_{w_2}(5) = \emptyset$. We have

$$\begin{aligned}\Delta(w, [\{3\}, \{2, 3\}]) &= (a_1 b_1 b_2 a_2 b_3, [\{3\}, \{2, 3\}]) \otimes (\mathcal{E}, [\emptyset, \emptyset]) \\ &\quad + (\mathcal{E}, [\emptyset, \emptyset]) \otimes (a_1 b_1 b_2 a_2 b_3, [\{3\}, \{2, 3\}]) \\ &\quad + (a_1 b_1 b_2, [\{3\}, \{2, 3\}]) \otimes (b_1 a_2 b_3, [\emptyset, \{1\}]) \\ &\quad + (b_1 a_2 b_3, [\emptyset, \{1\}]) \otimes (a_1 b_1 b_2, [\{3\}, \{2, 3\}]).\end{aligned}$$

For any $(w, [A, B]) \in \mathcal{B}$, we define the counit on $\hat{\epsilon}_\Delta(w, [A, B]) = 1$ if $w = \mathcal{E}$ and $\hat{\epsilon}_\Delta(w, [A, B]) = 0$ if w is non-empty.

Definition

A pair of bialgebras in cointeraction is a pair (A, m_A, Δ) , and (B, m_B, δ) of bialgebras with a (right) coaction $\rho : A \rightarrow A \otimes B$ of B on A such that the product, coproduct, counit, and unit of A are morphisms of right comodules. In other words,

- $\rho(\mathcal{E}_A) = \mathcal{E}_A \otimes \mathcal{E}_B$ and for any $a, b \in A$, $\rho(ab) = \rho(a)\rho(b)$: ρ is an algebra morphism.
- $(\Delta \otimes \text{Id}) \circ \rho = m_{1,3,24} \circ (\rho \otimes \rho) \circ \Delta$ where $m_{1,3,24}$ is a linear map $m_{1,3,24} : A \otimes B \otimes A \otimes B \rightarrow A \otimes A \otimes B$ such that $m_{1,3,24}(a_1 \otimes b_1 \otimes a_2 \otimes b_2) = a_1 \otimes a_2 \otimes b_1 b_2$ for any $a_1, a_2 \in A$ and $b_1, b_2 \in B$.
- For any $a \in A$, $(\epsilon_A \otimes \text{Id}) \circ \rho(a) = \epsilon_A(a)\mathcal{E}_B$.

We can see that $m_{1,3,24} = (\text{Id} \otimes \text{Id} \otimes m) \otimes m_{1,3,2,4}$. Similar to our discussion on the algebra homomorphism, we have the transposition map $m_{1,3,2,4}$. The map $m_{1,3,24}$ is important in the definition of cointeraction. For any $a \in A$, we can see that $(\Delta \otimes \text{Id}) \circ \rho(a) \in A \otimes A \otimes B$ and $(\rho \otimes \rho) \circ \Delta(a) \in A \otimes B \otimes A \otimes B$. Therefore, the map $m_{1,3,24}$ is required to move the elements in A into the correct positions and multiply the elements in B together.

We can check whether the pair of bialgebras (Wl, m, Δ) and (Wl, m, δ) with $\rho = \delta$ are in cointeraction.

Since (Wl, m, Δ) is a bialgebra, the coproduct δ is an algebra homomorphism, so the first property is satisfied.

For any $(w, [A, B]) \in \mathcal{B}$, we have

$$(\epsilon_{\Delta} \otimes \text{Id}) \circ \delta(w, [A, B]) = \sum_{A \subset C \subset B} \epsilon_{\Delta}(w, [A, C]) \otimes (w, [C, B]),$$

so $(\epsilon_{\Delta} \otimes \text{Id}) \circ \delta(w, [A, B]) = 0$ if w is non-empty and

$(\epsilon_{\Delta} \otimes \text{Id}) \circ \delta(w, [A, B]) = (\mathcal{E}, [\emptyset, \emptyset])$ if $w = \mathcal{E}$. The right hand side is 0 if w is non-empty, and $(\mathcal{E}, [\emptyset, \emptyset])$ if $w = \mathcal{E}$, so the third property is satisfied.

Now we want to check the second the property. For any $(w, [A, B]) \in \mathcal{B}$, we have

$$\begin{aligned} & (\delta \otimes \delta) \circ \Delta(w, [A, B]) \\ = & \sum_{\substack{i \\ A_{w_{i,1}} \subset C \subset B_{w_{i,1}} \\ A_{w_{i,2}} \subset D \subset B_{w_{i,2}}}} (w_{i,1}, [A_{w_{i,1}}, C]) \otimes (w_{i,1}, [C, B_{w_{i,1}}]) \\ & \otimes (w_{i,2}, [A_{w_{i,2}}, D]) \otimes (w_{i,2}, [D, B_{w_{i,2}}]), \end{aligned}$$

so

$$\begin{aligned}
 & m_{1,3,24} \circ (\delta \otimes \delta) \circ \Delta(w, [A, B]) \\
 = & \sum_{\substack{i \\ A_{w_{i,1}} \subset \dot{C} \subset B_{w_{i,1}} \\ A_{w_{i,2}} \subset D \subset B_{w_{i,2}}}} (w_{i,1}, [A_{w_{i,1}}, C]) \otimes (w_{i,2}, [A_{w_{i,2}}, D]) \\
 & \otimes (\overline{w_{i,1} w_{i,2}}, [C \cup D^*, B_{w_{i,1}} \cup B_{w_{i,2}}^*]),
 \end{aligned}$$

and we have

$$\begin{aligned}
 & (\Delta \otimes \text{Id}) \circ \delta(w, [A, B]) \\
 = & \sum_{\substack{i \\ A \subset \dot{E} \subset B}} (w_{i,1}, [A_{w_{i,1}}, E_{w_{i,1}}]) \otimes (w_{i,2}, [A_{w_{i,2}}, E_{w_{i,2}}]) \otimes (w, [E, B]),
 \end{aligned}$$

where $\Delta(w) = \sum_i w_{i,1} \otimes w_{i,2}$.

Given $E_{w_{i,1}}$ and $E_{w_{i,2}}$, we can choose $C = E_{w_{i,1}}$ and $D = E_{w_{i,2}}$, such that the first two factors in the tensor product are equal.

Given C and D , we can choose E such that $E_{w_{i,1}} = C$ and $E_{w_{i,2}} = D$ such that the first two factors in the tensor product are equal. However, in the third factor of the tensor product, $\overline{w_{i,1}w_{i,2}}$ is not necessarily equal to w . For example, Let $w = a_1b_1a_2$, where $\tilde{f}_w(1) = \tilde{f}_w(3) = \{2\}$ and $\tilde{f}_w(2) = \emptyset$. We have $\Delta(w) = a_1b_1a_2 \otimes \mathcal{E} + \mathcal{E} \otimes a_1b_1a_2 + a_1b_1 \otimes b_1a_2 + b_1a_2 \otimes a_1b_1$, but $a_1b_1b_1a_2 \neq a_1b_1a_2 \neq b_1a_2a_1b_1$.

In fact, if $w \in \mathcal{W}_f$ has two or more different letters in Ω , we will always have summand $w_1 \otimes w_2$ in the coproduct where w_1 and w_2 each have at least one letter in Ω and $w_1 \neq w_2$, and then $\overline{w_1w_2} = \overline{w_2w_1}$. Therefore, the pair of bialgebras (Wl, m, Δ) and (Wl, m, δ) is not in cointeraction.

The cause of this issue is that when the deshuffle operation in the coproduct takes apart a word with respect to letters in Ω , we lose the information about how the parts fit together, so the multiplication map fails to reconstruct the original word. There are two ways in which we can resolve this issue, either by introducing an equivalence relation such that the multiplication map gives a word that is equivalent to the original word, or by introducing global information such that the multiplication map is able to give back the original word.

Let us consider the first approach. We introduce the irreducible decomposition D such that $D(\mathcal{E}, [\emptyset, \emptyset]) = (\mathcal{E}, [\emptyset, \emptyset])$ and for any $(w, [A, B]) \in \mathcal{B}$ with n letters in Ω , where $n \in \mathbb{Z}_+$, $D((w, [A, B]))$ returns a tensor product in $\bigotimes_{i=1}^n WI$ equivalent up to permutation of factors, where each factor $(w_i, [A_{w_i}, B_{w_i}])$, which we call a singleton, contains only one letter in Ω in w_i for any $i = 1, \dots, n$. These singletons are obtained by recursively applying deshuffling coproduct to each factor of the tensor product until there is only primitive part. By the universal property of vector spaces, the definition of D can be extended to WI . For example, let $w = a_1 b_1 b_2 a_2 b_3 a_3$ where $\tilde{f}_w(1) = \{2, 3\}$, $\tilde{f}_w(4) = \{2, 5\}$, $\tilde{f}_w(6) = \{3\}$, and $\tilde{f}_{w_2}(2) = \tilde{f}_{w_2}(3) = \tilde{f}_{w_2}(5) = \emptyset$. We have

$$\begin{aligned} & D(a_1 b_1 b_2 a_2 b_3 a_3, [\{3\}, \{2, 3\}]) \\ &= (a_1 b_1 b_2, [\{3\}, \{2, 3\}]) \otimes (b_1 a_2 b_3, [\emptyset, \{1\}]) \otimes (b_2 a_3, [\{1\}, \{1\}]). \end{aligned}$$

Note that the irreducible decomposition D is similar to the prime factorization of integers.

We define the equivalence relation $\sim$ such that for any $a, b \in WI$, $a \sim b$ if $D(a) = D(b)$.

Issues and Solutions IV

The second approach is to give a global order when we define the multiplication. We want to introduce an additional restriction that for any $(w, [A, B]) \in \mathcal{B}$ each letter in Ω in w appear only once. We introduce a global order $w_g \in \mathcal{W}_f$ in which each letter in the alphabet Ω appears exactly once. We also have the global association map $\tilde{f}_{w_g}$ that associates letters in Ω' in w_g with letters in Ω . Then all the valid words are formed by taking some letters in Ω from w_g and filling in letters from Ω' in a way that preserves the correspondence relation and the relative position. We keep the definitions of $\Delta, \delta, \epsilon_\Delta, \epsilon_\delta$ and u and change the multiplication map m . For any $(w_1, [A, B]), (w_2, [C, D]) \in RWI$ (RWI is restricted WI in the way described above), let

$$m((w_1, [A, B]) \otimes (w_2, [C, D])) = (w_1 \cup w_2, [(A \cup C)^\dagger, (B \cup D)^\dagger]),$$

where $w_1 \cup w_2$ is the word formed by taking the union of letters in Ω in w_1 and w_2 and filling in letters from Ω' in a way that preserves the correspondence relation and the relative position with

respect to w_g , and $(A \cup C)^\dagger$ (or similarly $(B \cup D)^\dagger$) is the set of indices of letters in Ω' pointed to by A and C adjusted with respect to the new word $w_1 \cup w_2$. Following the similar proofs as before, we can show that (RWI, m, Δ) and (RWI, m, δ) are bialgebras. For example, consider the graph G , and let

$$w_g = a_1 b_1 b_2 b_4 a_3 a_2 b_3 a_4,$$

and $\tilde{f}_{w_g}(1) = \{2, 3, 4\}$, $\tilde{f}_{w_g}(5) = \{2, 3\}$, $\tilde{f}_{w_g}(6) = \{4, 7\}$, $\tilde{f}_{w_g}(8) = \{3, 7\}$, and $\tilde{f}_{w_g}(2) = \tilde{f}_{w_g}(3) = \tilde{f}_{w_g}(4) = \tilde{f}_{w_g}(7)$. We have $m((a_1 b_1 b_2 b_4 a_2 b_3, [\{2\}, \{2, 3\}]) \otimes (b_1 b_2 b_4 a_3 a_2 b_3, [\{6\}, \{1, 6\}])) = (a_1 b_1 b_2 b_4 a_3 a_2 b_3, [\{2, 7\}, \{2, 3, 7\}])$.

Pros of the first approach: extending the definition from WI to the quotient space is natural.

Cons of the first approach: totally annihilates the order structure in the graph.

Pros of the second approach: keeps the order structure in the graph.

Cons of the second approach: the global ordering can be extremely long if Ω has many elements and there is more restriction on the vector space RWI compared to WI .